THROTTLE POSITION (TP) SENSOR [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

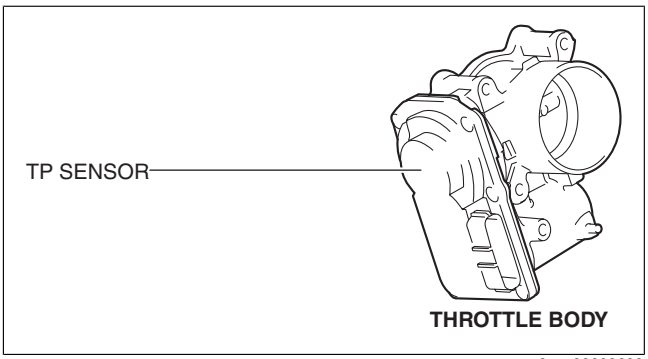
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Purpose/Function

- Detects the throttle valve opening/closing condition, which operates in accordance with the accelerator pedal operation, as basic information (mainly used to determine the fuel injection amount) and sends it to the PCM as the throttle valve opening angle signal.

Construction

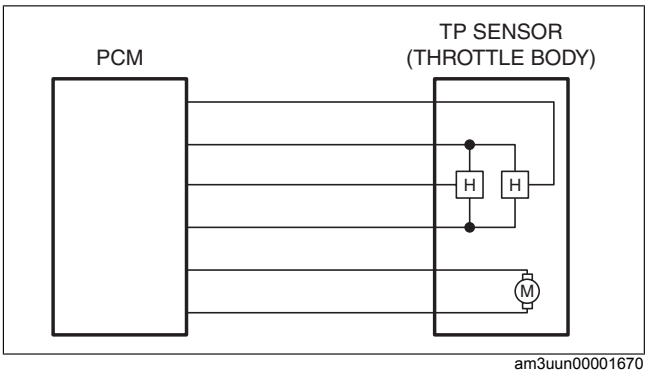
- Built into the throttle body.
- A Hall element has been adopted (as the detecting element) for detecting the throttle valve opening/closing condition.
- The TP sensor has a main sensor (No.1) and a sub sensor (No.2) for improved detection accuracy.



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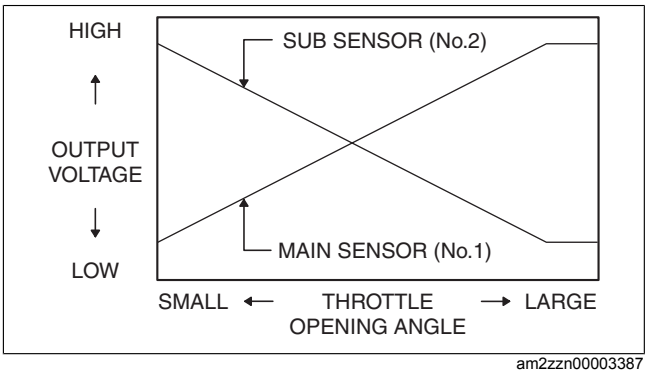
Operation

- Sends the signal, which fluctuates according to the throttle valve opening/closing, to the PCM.



am3uun00001670

TP sensor characteristic



am2zzn00003387

Fail-safe

DTC No.	Fail-safe function
P0122:00	• Restricts the upper limit of the engine speed.
P0123:00	• Restricts the upper limit of the engine speed.
P0222:00	• Restricts the upper limit of the engine speed.
P0223:00	• Restricts the upper limit of the engine speed.
P2101:00	• Restricts the upper limit of the engine speed. • Stops the drive-by-wire control (throttle valve is open at approx. 8° by return spring force).
P2109:00	• Restricts the upper limit of the engine speed. • Stops the drive-by-wire control (throttle valve is open at approx. 8° by return spring force).
P2135:00	• Restricts the upper limit of the engine speed. • Stops the drive-by-wire control (throttle valve is open at approx. 8° by return spring force).